

SENG1050: Web Technologies

Callaghan

Semester 2 - 2025



THE UNIVERSITY OF
NEWCASTLE
AUSTRALIA

OVERVIEW

Course Description

This course introduces web technologies and the fundamental concepts of internet architecture and how they support the massive growth and varied uses of the medium. A strong emphasis is placed on practical skills such as using various communication techniques, building web pages, and securing information via encryption. The course is designed to give students a sound understanding of the potential as well as the limitations of web technology. Previous computing experience is helpful but not required.

Academic Progress Requirements

Nil

Contact Hours

Callaghan Computer Lab *

Face to Face On Campus

2 hour(s) per week(s) for 11 week(s) starting Week 2

Lecture

Face to Face On Campus

2 hour(s) per week(s) for 13 week(s) starting Week 1

Unit Weighting Workload

* This contact type has a compulsory requirement.

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Students are required to spend on average 120-140 hours of effort (contact and non-contact) including assessments per 10 unit course.

COURSE OUTLINE

www.newcastle.edu.au

CRICOS Provider 00109J

CONTACTS

Course Coordinator **Callaghan**
Dr Adrian Tan
Adrian.Tan10@newcastle.edu.au
(02) 4055 0700
Consultation: Thursdays 10 am to 12 noon (or other times by appointment) at SR211

Teaching Staff Other teaching staff will be advised on the course Canvas site.

School Office **School of Information and Physical Sciences**
SR233, Social Sciences Building
Callaghan
CESE-SIPS-Admin@newcastle.edu.au
+61 2 4921 5538
9am-5pm (Mon-Fri)

SYLLABUS

Course Content

- Introduction to computer networking
- Past, present and potential future of the Internet
- Client-server computing, Search engines, Internet protocols
- World-wide Web Languages: HTML, XML, CSS and Javascript
- HCI and communication issues relating to Web page design
- Cryptography
- Compression
- Social Aspects of the Internet: privacy, responsibility and legal issues.

Course Learning Outcomes **On successful completion of this course, students will be able to:**
1. Sound understanding of basic web technology architectures

2. Familiarization with a variety of aspects of electronic communication.

3. Understanding of syntax and semantics of several Markup languages for use in information communication.

4. Detailed knowledge of the concepts and practical aspects of data encryption and compression in networking environments.

Course Materials **Recommended Text:**
- Web Development and Design Foundations with HTML5 (9e) by Terry Felke-Morris, Pearson

- Data Communications and Networking With TCP/IP Protocol Suite (6e) by Behrouz A. Forouzan, McGraw-Hill

- Programming the World Wide Web (8e) by Robert W. Sebesta, Pearson

COMPULSORY REQUIREMENTS

In order to pass this course, each student must complete ALL of the following compulsory requirements:

Contact Hour Requirements:

- Computer Lab Attend 80% of sessions

Course Assessment Requirements:

- Assessment 4 - Final exam: Pass requirement 40% - Must obtain 40% in this assessment item to pass the course.

Compulsory Placement and WHS Requirements:

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ASSESSMENTS

This course has 4 assessments. Each assessment is described in more detail in the sections below.

	Assessment Name	Due Date	Involvement	Weighting	Learning Outcomes
1	Mid-semester quiz	Week 7 (12 Sep 2025, 23:59 pm)	Individual	20%	1, 2
2	Web-based assignment 1	Week 6 (05 Sep 2025, 23:59 pm)	Individual	15%	3
3	Web-based assignment 2	Milestone 1: Checked during Week 8 labs (15 Sep 2025 - 19 Sep 2025) Milestone 2: Checked during Week 10 labs (06 Oct 2025 - 10 Oct 2025) Assessment due: Week 12 (24 Oct 2025, 23:59 pm)	Individual	25%	3
4	Final exam*	Formal examination period	Individual	40%	1, 2, 3, 4

* This assessment has a compulsory requirement.

Late Submissions

The mark for an assessment item submitted after the designated time on the due date, without an approved extension of time, will be reduced by up to 10% of the possible maximum mark for that assessment item for each day or part day that the assessment item is late. Note: this applies equally to week and weekend days.

Assessment 1 - Mid-semester quiz

Assessment Type

Quiz

Purpose

The purpose and benefit of the quiz is to provide students with mid-term feedback on student learning. This quiz will highlight areas of concern and may stimulate discussion with tutors and lecturers.

Description

Mixed response quiz.

Weighting

20%

Due Date

Week 7 (12 Sep 2025, 23:59 pm)

Submission Method

Online

Assessment Criteria

Refer to Canvas.

Return Method

Online

Feedback Provided

Online

Opportunity to

Students will not be given the opportunity to reattempt this assessment.

Reattempt

Assessment 2 - Web-based assignment 1

Assessment Type

Project

Purpose

To author an initial web page using HTML and CSS. To create an elementary XML document with accompanying DTD.

Description

Individual assignment.

Weighting	15%
Due Date	Week 6 (05 Sep 2025, 23:59 pm)
Submission Method	Online
Assessment Criteria	Refer to Canvas.
Return Method	Online
Feedback Provided	Online
Opportunity to Reattempt	Students will not be given the opportunity to reattempt this assessment.

Assessment 3 - Web-based assignment 2

Assessment Type	Project
Purpose	To extend from Assessment 2 (Web-based Assignment 1) and complete the website by including advanced HTML, XML, and Javascript.
Description	Individual assignment.
Weighting	25%
Due Date	Milestone 1: Checked during Week 8 labs (15 Sep 2025 - 19 Sep 2025) Milestone 2: Checked during Week 10 labs (06 Oct 2025 - 10 Oct 2025) Assessment due: Week 12 (24 Oct 2025, 23:59 pm)
Submission Method	Online
Assessment Criteria	Refer to Canvas.
Return Method	Online
Feedback Provided	Online
Opportunity to Reattempt	Students will not be given the opportunity to reattempt this assessment.

Assessment 4 - Final exam

Assessment Type	Formal Examination
Purpose	The final formal examination is designed to test the individual student's knowledge of the course material and their ability to describe, analyse and hypothesise from this material.
Description	Formal examination.
Weighting	40%
Compulsory Requirements	Pass requirement 40% - Must obtain 40% in this assessment item to pass the course..
Due Date	Formal examination period
Submission Method	Formal Exam
Assessment Criteria	
Return Method	Not Returned
Feedback Provided	No Feedback
Opportunity to Reattempt	Students will be given the opportunity to reattempt this assessment.

ADDITIONAL INFORMATION

Grading Scheme

This course is graded as follows:

Range of Marks	Grade	Description
85-100	High Distinction (HD)	Outstanding standard indicating comprehensive knowledge and understanding of the relevant materials; demonstration of an outstanding level of academic achievement; mastery of skills*; and achievement of all assessment objectives.
75-84	Distinction (D)	Excellent standard indicating a very high level of knowledge and understanding of the relevant materials; demonstration of a very high level of academic ability; sound development of skills*; and achievement of all assessment objectives.
65-74	Credit (C)	Good standard indicating a high level of knowledge and understanding of the relevant materials; demonstration of a high level of academic achievement; reasonable development of skills*; and achievement of all learning outcomes.
50-64	Pass	Satisfactory standard indicating an adequate knowledge and

	(P)	understanding of the relevant materials; demonstration of an adequate level of academic achievement; satisfactory development of skills*; and achievement of all learning outcomes.
0-49	Fail (FF)	Failure to satisfactorily achieve learning outcomes. If all compulsory course components are not completed the mark will be zero. A fail grade may also be awarded following disciplinary action.

Attendance

*Skills are those identified for the purposes of assessment task(s).

Attendance/participation will be recorded in the following components:

- Computer Lab (Method of recording: Manual attendance recording)

Communication Methods

Communication methods used in this course include:

- Canvas Course Site: Students will receive communications via the posting of content or announcements on the Canvas course site.
- Email: Students will receive communications via their student email account.
- Face to Face: Communication will be provided via face to face meetings or supervision.

Course Evaluation

Each year feedback is sought from students and other stakeholders about the courses offered in the University for the purposes of identifying areas of excellence and potential improvement.

Learning Analytics

The University uses an evidence-based approach to enhancing student learning and delivering support. Our careful analysis of data through learning analytics informs decision making processes related to student learning, academic outcomes, and support services. Learning analytics may be used to identify targeted individual opportunities for additional support services or interventions.

Interviews on Assessment Items

An Interview may be conducted on any assessment item in this course, in accordance with the [Interview on Assessment Items Procedure](#). The purpose of the interview is to verify the author of the material submitted in response to the assessment task and provide a quality assurance measure.

In the event the Course Coordinator is not satisfied that a student's oral responses are consistent with the work originally submitted, the matter will be referred to the Student Academic Conduct Officer, who will take appropriate action under the [Student Conduct Rule](#).

Academic Integrity and Ethical Academic Conduct Policy

All students are required to meet the academic integrity standards of the University. These standards reinforce the importance of integrity and honesty in an academic environment. Academic Integrity policies apply to all students of the University in all modes of study and in all locations.

Please refer to the [Academic Integrity and Ethical Academic Conduct Policy](#)

Adverse Circumstances

The University acknowledges the right of students to seek consideration for the impact of allowable adverse circumstances that may affect their performance in assessment item(s).

Applications for special consideration due to adverse circumstances will be made using the online Adverse Circumstances system where:

1. the assessment item is a major assessment item; or
2. the assessment item is a minor assessment item and the Course Co-ordinator has specified in the Course Outline that students may apply the online Adverse Circumstances system;
3. you are requesting a change of placement; or
4. the course has a compulsory attendance requirement.

Before applying you must refer to the [Adverse Circumstance Affecting Assessment Items Procedure](#).

Use of generative artificial intelligence in course assessments

It is critical that any work submitted for assessment is your own original work. Before using generative AI tools (such as ChatGPT, Perplexity, Microsoft Copilot, etc) in any assessable work you must ensure that such use is in line with the requirements for the course and expectations of your Course Coordinator

Misuse of AI tools may be considered a breach of the [University's Student Conduct Rule](#) and could result in disciplinary action.

Artificial Intelligence detection software may be used to review any work you submit. If you have used AI in any way other than has been expressly permitted by your course coordinator, you may be engaging in academic misconduct and be subject to penalties.

For information, refer to:

- [Generative AI Tools](#)
- [Academic Integrity](#)

Important Policy Information

The Help button in the Canvas Navigation menu contains helpful information for using the Learning Management System. Students should familiarise themselves with the policies and procedures at <https://www.newcastle.edu.au/current-students/respect-at-uni/policies-and-procedures> that support a safe and respectful environment at the University

GRADUATE PROFILE STATEMENTS

The following table illustrates how this course contributes towards building the skills students will need to work in their profession.

Level of capability

- Level 1 indicates an introduction to a topic at a university level
- Levels 2 and 3 indicate progressive reinforcement of that topic
- Level 4 indicates skills commensurate with a graduate – entry to professional practice
- Level 5 indicates highly specialist or professional ability

Bachelor of Computer Science

	University of Newcastle Bachelor of Computer Science Graduate Profile Statements	Taught	Practised	Assessed	Level of capability
1	Knowledge of basic science and computer science fundamentals.				
2	In depth technical competence in the discipline of computer science.	X	X	X	1
3	An ability to carry out problem analysis, requirements capture, problem formulation and integrated software development for the solution of a problem.				
4	Capacity to continue developing relevant knowledge, skills and expertise in computer science throughout their careers.	X	X	X	1
5	An ability to communicate effectively with other Computer Scientists, Software Engineers, other professional disciplines, managers and the community generally.	X	X	X	1
6	Ability to undertake and co-ordinate large computer science projects and to identify problems, their formulation and solution.				
7	Ability to function effectively as an individual, a team member in multidisciplinary and multicultural teams and as leader/manager with capacity to assist and encourage those under their direction.				
8	Understanding of social, cultural, global and business opportunities of the professional computer scientist; understanding the need for and principles of sustainability and adaptability.				
9	Understanding of professional and ethical responsibilities and a commitment to them.				
10	Understanding of entrepreneurship; need of and process of innovation, as well as the need of and capacity for lifelong learning.				

Bachelor of Engineering

	University of Newcastle Bachelor of Engineering Graduate Profile Statements	Taught	Practised	Assessed	Level of capability
	Knowledge Base				
1	1.1. Comprehensive, theory based understanding of the underpinning natural and physical sciences and the engineering fundamentals applicable to the engineering discipline.				
2	1.2. Conceptual understanding of the mathematics, numerical analysis, statistics, and computer and information sciences which underpin the engineering discipline.				
3	1.3. In-depth understanding of specialist bodies of knowledge within the engineering discipline.	X	X	X	1
4	1.4. Discernment of knowledge development and research directions within the engineering discipline.	X	X	X	1
5	1.5. Knowledge of contextual factors impacting the engineering discipline.				
6	1.6. Understanding of the scope, principles, norms, accountabilities and bounds of contemporary engineering practice in the specific discipline.				
	Engineering Ability				
7	2.1. Application of established engineering methods to complex engineering problem solving.				
8	2.2. Fluent application of engineering techniques, tools and resources.	X	X	X	1
9	2.3. Application of systematic engineering synthesis and design processes.				
10	2.4. Application of systematic approaches to the conduct and management of engineering projects.				
	Professional Attributes				
11	3.1. Ethical conduct and professional accountability				
12	3.2. Effective oral and written communication in professional and lay domains.				
13	3.3. Creative, innovative and pro-active demeanour.				
14	3.4. Professional use and management of information.				
15	3.5. Orderly management of self, and professional conduct.				
16	3.6. Effective team membership and team leadership.				

Bachelor of Information Technology

	University of Newcastle Bachelor of Information Technology Graduate Profile Statements	Taught	Practised	Assessed	Level of capability
1	Demonstrate a comprehensive understanding of the discipline of information technologies with an emphasis on net-centric applications, information management, and user requirements for ethical professional practice.	X	X	X	1
2	Apply critical reasoning and systems thinking to understand and support the operation and constraints of contemporary enterprises and their dynamic environment.				
3	Work independently and collaboratively to locate, manage and organise information and resources and apply evidence-based methodologies to create, modify and maintain designs and design solutions.	X	X	X	1
4	Use creativity, problem solving skills, project management skills and technical expertise to analyse, interpret, evaluate and generate solutions to complex technical and organisational problems.				
5	Demonstrate professional judgement and responsibility by communicating information technology principles, practices, standards to specialist and non-specialist audience clearly and persuasively.	X	X	X	1

This course outline was approved by the Head of School. No alteration of this course outline is permitted without Head of School approval. If a change is approved, students will be notified and an amended course outline will be provided in the same manner as the original.

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